

aircraft and returning to a receiver, on the F-117 this was negated by using angled flat surfaces that deflected signals instead of bouncing them back. Beyond this, each element of the aircraft, from the cockpit to the tailfins to the engine air intakes, was designed to minimize radar reflection. The other route to stealth was to cover the airframe with radar-absorbent materials – paint and tiles that would transform the energy of the radar pulse into heat instead of reflecting it.

The F-117 depended absolutely on radar-invisibility for combat survival. It was subsonic, not especially maneuverable, and even lacked electronic countermeasures (ECM) equipment. If progress in radar technology or some other tracking method had been able to tear off the Nighthawk's cloak of invisibility, it would have been defenseless. But over Iraq it performed outstandingly – responsible for 30 percent of targets hit by precision munitions.

Precision bombing

Essential to the F-117's accuracy of attack was, of course, the improved quality of guidance systems for smart bombs and missiles. There was no essential difference between the munitions being employed in the Gulf War from those used against North Vietnam in the early 1970s, but the effectively presented images of precision attack fed to the media during Operation Desert Storm made a telling impression on the public imagination. And, although subsequent inquiry revealed that many weapons had fallen short of the effectiveness prematurely attributed to them during the war, the laser-guided, TV-guided, and antiradiation missiles and bombs were unquestionably more accurate than anything that had been seen in action before.

Perhaps the most impressive aspect of air



GROUND STRIKE

First delivered to the US Navy in 1981, the McDonnell-Douglas F/A-18 Hornet was, as its designation indicates, a dual-role aircraft – both an agile air-superiority fighter and a powerful ground-attack aircraft. During the Gulf War in 1991 the Hornets were mostly used in a ground-attack role.



DIRECT HIT

A laser-guided missile is launched at an ammunition depot in Iraq during the Gulf War in January 1991. This was the view of the event as seen through the sights of a French fighter.

power in the Gulf War was its effect on the land battlefield, where it contributed to a victory being achieved with the minimum number of Coalition casualties. The veteran B-52s were once more brought into action, dropping their impressive tonnage of iron bombs on the Iraqis dug in to lines of trenches in the desert. But the crucial damage was done by newer technology. Two Boeing-Northrop Grumman E-8 J-STARS (Joint Surveillance and Target Attack Radar System) aircraft did for the air-land battlefield what AWACS had done for the air-to-air battle. Like the AWACS E-3s, the E-8s were derived from the Boeing 707. J-STARS controllers were able to observe every move of the Iraqi army, directing strike aircraft to destroy armored columns, truck convoys, fuel dumps, or artillery emplacements. The airborne radars offered such high-resolution imagery that they could even pick out individual vehicles for attack.

The panoply of ground-attack aircraft at the Coalition's disposal ranged from AH-64 helicopter gunships to A-10 Thunderbolt II "Warthog" tank-busters and F-16s or F-18s in ground-attack mode. Linked to J-STARS, this aerial array was deadly. For example, on January 22, 1991, 60 Iraqi tanks were spotted on the move; air strikes were called in and 58 of the vehicles were destroyed. The carnage of the attempted Iraqi retreat from Kuwait in February was also orchestrated by J-STARS, monitoring convoys crowding the road back to Iraq and calling in wave after wave of F-18s.

Victory by air power

The events of the Gulf War were a challenge to the accepted wisdom – based on the experience of World War II, Korea, and Vietnam – that air power alone could not win a war. USAF Chief of Staff General Merrill A. McPeak asserted bluntly that Desert Storm marked "the first time in air history that a field army has been defeated by air power." There were particular reasons why aircraft were at their maximum effectiveness in this conflict. Iraq presented a range of targets suitable for strategic air attack, from command and control centres to weapons factories, missile sites, and airfields. Similarly, in the ground war, troops in fixed emplacements and tanks moving in open desert terrain both offered

SKY SENTRY

The E-3 Sentry early-warning and control aircraft – essentially a Boeing 707 with a saucer-shaped radar antenna on its back – was the hub of Coalition air operations in the Gulf War.

